# Effect of Finger and Hand Exercises among Amateur Pianist

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## Abstract

**BACKGROUND**: Now days the number of students studying piano in tertiary institution is being increasing. The piano player is similar to athletic activity to play the notes accurately. Posture plays a vital role in piano. Musicians are prone to injuries due to the nature of musical practice, inappropriate body postures and potentially harmful playing techniques. Majority of amateur players are prone to injuries .Amateur piano player experience pain in hands and fingers. The main objective of the study was to strengthen the finger and hand muscles to reduce the level of pain. **Methodology**: Quasi Experimental study and convenient sampling.20 amateur piano players, were selected according to inclusion and exclusion criteria. Pre test was done to the participants using VAS score, hand grip dynamometer and cornell musculoskeletal discomfort questionnaire. Finger and hand Exercise were given to the participants for 4 weeks . After 4 weeks of study to analyze their level of pain, a hand held cornell musculoskeletal discomfort questionnaire, VAS score and hand grip dynamometer were used to find the effect of finger and hand exercise exercise. **Outcome Measure:** CORNELL MUSCULOSKELETAL DISCOMFORT QUESTIONNIARE, VAS SCORE,HAND GRIP DYNAMOMETER **Result** & **Conclusion:** The study concludes that there was significant reduction in the level of pain and also the hand grip was significantly improved

***Keywords:*** *Amateur Piano player, harmful playing techniques, finger and hand pain,*

## Introduction

Piano is a classical and fascinating instrument. In 18th century Bartolomeo Christofori invented first generation of piano and twentieth century piano was highly developed and its quality , both in mechanism and sound reached its peak .Playing Piano notes accurately is similar to the athletic activity. The touch of each key weights around 52g for present day acoustic piano. Piano needs an application of greater muscle effort to stabilize the finger joints[1].

Most of the public music is often perceived as fun and safe job without posing any harm to the body .There is a possibility for musicians to suffer from injuries. Pianist are prone injuries due to nature of musical practice. The occurrence of "Playing related musculoskeletal disorders" is almost unavoidable when a given motion is highly repetitive when combined with the prolonged use of body segments and without a proper understanding of the anatomical limits and motion ranges of the Human body.

From the history of piano playing, some experience regarding injuries was recorded such as Robert and Clara Schumann and Sergei Rachmaninov. In twentieth century when piano was highly developed and its quality both in mechanism and sound reached at its peak, many contemporary pianists reported their symptoms and their horrible experience after injuries were developed such as Glenn Gould, and Leon Fleisher (Mark, 1999). In most of cases, injuries can be so severe that they cannot even perform and play some kind of repertoire[3].

The present research has shown that Prevalence of PRMD has been reported in the range of 25%to 93%.[20] Now days students learning music in tertiary institution

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has been increasing. The students require extreme mental and physical endurance strength to perform. Although piano techniques vary in different genres of music, examples of faulty piano techniques that could put pianists at risk of PRMD include fingers pressing too hard on the keyboard, head deviating from the central position, elevated shoulder, curving of the wrist which should be in a straight line with the fifth finger, forearm raised until it is not parallel with the floor, elbow too close or too far from the body, and locked wrist[7,21]

A regular practice session on daily basis is required for a pianist to reach the required standard .Some piano pieces require more grip and strength in fingers, for example pieces like "MODERATO''AND ALLEGRO .So to play the notes accurately the students practice a lot , A sudden increase of practice time before examination and performance is one such cause of PRMD. Lack of finger exercise prior to and after piano playing is also a risk factor of PRMD . As well, a lack of warm-up and cool-down exercises prior to and afterpiano playing is also a risk factor for PRMD[21].The main objective of the study is to strengthen the finger and hand muscles and to reduce the level of pain.

## Materials and Methods

The study is an Quazi Experimental study which was conducted in 2020 at Kingsway Music Academy . In this study twenty amateur piano students of both men and women between the age 18 -21 were recruited. The Grade exam appearing students who practice for 60 mins a day and who had Playing related musculoskeletal disorder in Hand and wrist were recruited. The students who learn other instrument along with piano were excluded. Initially the level of pain and hand grip was measure using Vas Score and Hand grip Dynamometer and Cornell musculoskeletal discomfort questionnaire was given to disclose the discomfort in the hand among the participants.

The exercise protocol had been charted out , for 4 weeks duration, twice a day, weekly thrice, 3sets,10 times repetition. Stretching exercise for forearm and finger flexors, lumbricals and thumb extensors with 5 seconds hold. Free exercises such as finger opening and closing, finger taps and thumb bending exercises were charted for the participants. Elastic band resistance was used to strengthen the finger and thumb abductors. Digi flex finger strengthener was used for strengthening MCP and interphalangeal joint flexor.

Then after 4 weeks of exercise, the Vas score was measured to analyze the level of pain and Hand grip dynamometer to measure the hand grip.Cornell musculoskeletal discomfort questionnaire was given to disclose the discomfort in the hand among the participants.

## Result

A total 20 participants were selected in which 11 were women and 9 were men .Out of 20 students 25% were 18 years of age ,25 % were 19 years of age ,25% of them 20 years of age and 25% of them were 21 years of age.

From Table III the pre and post test value, analyzed according to Cornell musculoskeletal discomfort questionnaire. In Area A the mean value of pre test was 5.05 and post test was 2.750. In Area B mean value of pre test was 9.05 and post test was 4.925. In Area C mean value of pre test value was 2.55 and post test value was 2.125. In Area D the mean value of pre test value was 0.50 and post test value was 0.250.In Ares E the mean value of pre test value was 3.20 and post test value was 1.8255 .In Area F the mean value of pre test value was 4.50 and post test value was 2.050. The pre and post test values analyzed using questionnaire show that pain has been significantly decreased

From Table IV the level of pain and hand grip was analyzed. The mean value of vas score in pre test was 5.15 and post test was 3.750 in which the level of pain had been significantly decreased The mean value of hand grip in pre test was 10.25 and post test was 12.500 in which hand grip has been increased.

**Table I: Age Frequency**

| Age | Frequency | Percentage |
|---|---|---|
| 18 | 5 | 25.0 |
| 19 | 5 | 25.0 |
| 20 | 5 | 25.0 |
| 21 | 5 | 25.0 |
| Total | 20 | 100.0 |

**TABLE II: GENDER FREQUENCY**

| GENDER | FREQUENCY | PERCENTAGE |
|---|---|---|
| FEMALE | 11 | 55.0 |
| MALE | 9 | 45.0 |

**TABLE III: CORNELL MUSCULOSKELETAL DISCOMFORT QUESTIONNAIRE (PRE & POST TEST)**

| | | Mean | N | Std. Deviation | t – Value | P - Value |
|---|---|---|---|---|---|---|
| Area A | Pre | 5.05 | 20 | 4.186 | 3.467 | 0.003 S |
| | Post | 2.750 | 20 | 3.0414 | | |
| Area B | Pre | 9.05 | 20 | 2.350 | 5.133 | 0.000 S |
| | Post | 4.925 | 20 | 3.3611 | | |
| Area C | Pre | 2.55 | 20 | 3.720 | 0.838 | 0.412 NS |
| | Post | 2.125 | 20 | 3.2842 | | |
| Area D | Pre | 0.50 | 20 | 2.236 | 1.000 | 0.330 NS |
| | Post | 0.250 | 20 | 1.1180 | | |
| Area E | Pre | 3.20 | 20 | 4.895 | 1.729 | 0.100 NS |
| | Post | 1.825 | 20 | 3.3296 | | |
| Area F | Pre | 4.50 | 20 | 4.560 | 3.621 | 0.002 S |
| | Post | 2.050 | 20 | 2.2936 | | |

**TABLE IV : VAS AND HAND GRIP (PRE & POST TEST)**

| | | Mean | N | Std. Deviation | t – Value | P - Value |
|---|---|---|---|---|---|---|
| VAS | Pre | 5.15 | 20 | .671 | 6.294 | 0.000 S |
| | Post | 3.750 | 20 | .9665 | | |
| HAND GRIP | Pre | 10.25 | 20 | 2.552 | -3.943 | 0.001 S |
| | Post | 12.500 | 20 | 3.8044 | | |

## Discussion

Piano, the hammer incorporated in it strikes the strings and reproduces the sound of musical note, when the key pressed stronger,the velocity of the hammer striking the strings is increased and more sound will be produced[1] this 20th century many young population particularly school going students are interested to learn playing piano as an recreational activity and as an passion on music.

The musculoskeletal injuries in work and recreational activities like sports are common, similarly playing musical instruments is also prone to cause musculoskeletal injuries called Playing-Related Musculoskeletal Disorders (PRMDs).

This PRMDs is common among amateur piano players, mainly due to the posture which assumed and maintained during playing the piano,the wrist and hand symptoms are common among them. There are three types of wrist posture commonly assumed during playing piano, one is high wrist, neutral wrist and weight piano playing. The first one high wrist is preferred by few piano players which is traditional, in this wrist will be higher than the keyboard, the second one is neutral wrist in this the piano player assumes the wrist in neutral posture, the wrist corresponds appropriately to the level of keyboard, the third one weight on piano in this wrist rotates while playing and weight of the hand and wrist will be on the fingers. While playing piano along with wrist and fingers, the elbow and shoulder joints is also involved. While playing piano the posture assumed by the fingers can be flat or rounded, posture of elbow can be straight or bent and posture of shoulder can be elevated or non-elevated. The motion of the fingers can horizontal or vertical while playing piano. Apart from the joints of upper extremities, there will be involvement of motion of neck and torso while playing piano([2])

The wrist joint connects the hand an sensory organ to upper extremity. Numerous muscles,tendons,nerves and vascular structures of hand transverses wrist joint. The neutral wrist joint posture plays a crucial role for normal functioning of these structures,which tranversing it and in preventing injuries to these structures from mechanical origin.

The commonly assumed wrist postures during playing the piano are high wrist,neutral wrist and weight of piano. These postures are not an neutral posture for wrist joint, the neutral posture of wrist is forearm in mid prone postion,wrist joint neutral with out any flexion,extension,radial and ulnar deviation([5])

In all the three postures which commonly assumed during playing piano, the forearm will be in a fully pronated position, in which radius shaft will be twisted on ulna and with arthro kinematic changes in the inferior radio-ulnar joint such as radius will be migrates medially and superiorly on ulna and ulna abducts and migrates posteriorly(4).

In high wrist posture and weight on piano playing the wrist will be in flexion, the proximal carpal row of wrist will moves volarly and distal row palmarly(4)

In all the three postures of wrist while playing piano, in the MCP joints of fingers, the proximal phalanx will slides and rolls palmarly on flexion and volarly on extension, the proximal phalanx and slides and rolls laterally during abduction, in the corresponding metacarpals. In the interphalangeal joints of fingers,the phalanx will be rolling and sliding palmarly and volarly on finger flexion and extension movements.The first metacarpal of thumb slides and rolls on trapezium palmarly on abduction,medially on flexion,laterally on extension.(3)(6)

The repetitive motion and sustained posture of wrist while playing piano and the arthro-kinematic changes due to it, in inferior radio-ulna joint,carpal joints,MCP joints and interphalangeal joints interfere with structures transversing the joint, commonly there will be length-tension changes in the musculo-tendinous structures and it is prone to cause muscle fatigue,myofascial trigger points,tendinitis and etc. Most commonly the long flexors and extensors of wrist and fingers,thenar muscles,lumbricals,palmar and dorsal introssei muscles are involved. At times the neural structures adjacent to these musculo-skeletal structures is also involved in causing pain and numbness.

The wrist posture other than the neutral will over tax the musculo-tendinous structures and it is prone to cause PRMDs and loss of grip strength(7). Same time the repetitive motion of the fingers and thumb are prone to PRMDs. The long flexors and extensors of wrist and fingers are affected by non-neutral wrist posture and repetitive motions, the intrinsic musculature are affected by sustained finger flexion and abduction and repetitive motions.

The muscular structures of wrist and hand are primarily involved in causing PRMDs among piano players. The PRMDs is more prevalent among adult and children amateur piano players. The common cause of the PRMDs among the amateur players are improper posture of wrist assumed during practice and playing,

lack of strength and flexibility in wrist and finger muscles.

In this study 20 amateur piano player,both male and female genders, between the age group of 18 to 21 years ,who had PRMDs in wrist and hand due to playing piano had been selected in and around chennai. In this study on pre-test The pain had been measured with VAS scale,the hand grip strength were measured with hand held dynamometer and cornell musculoskeletal discomfort questionnaire was given to disclose the discomfort in the hand among the participants.

The exercise protocol had been charted out for this 20 participants, for 4 weeks duration,twice a day,weekly thrice, 3sets,10 times repetition. Stretching exercise for forearm and finger flexors,lumbricals and thumb extensors with 5 seconds hold. Free exercises such as finger opening and closing,finger taps and thumb bending. Exercise with elastic resistance band to strengthen finger and thumb abductors and finger play strengthener for strengthening MCP and interhalangeal joint flexors.After 4 weeks chartered exercise protocol, there was significant improvement among this 20 piano players pain and discomfort in wrist and hand.

## Conclusion

The study concludes that lack of warm up and cool down exercise among amature pianist is the cause of PRMD**.** The study also concludes that Correct positioning at the piano can play a large role in achieving correct balance and posture creating tension free playing. The possible prevention of future occurrence of playing related musculoskeletal disorder is to seek a set of suitable playing technique that is user friendly. Along with posture warm up and cool down exercise can be charted for the amateur piano players so that level finger pain ,and fatigue can be minimized.

**Limitation**

Sample size is less.

Limited age group was included.

Protocol Duration was lesser.

**Recommendation**

Interference with large sample.

Exercise Duration could be increased.

Motion capture analysis camera can be used

Appropriate Questionnaire can be designed

**Conflict of Interest**: Nil

**Source of Funding: Self Funding**

**Ethical Clearance:** Institutional Ethical Committe

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